

LEARNING CANON

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Abstract

LEARNING is INTEL applied governed discovery extraction at scale, the IDF generalized beyond patents to every scope.

hadleylab.org Governed Research. Every claim cited.

Contents

- **LEARNING_IS_INTEL_APPLIED** INTEL is the primitive (what you KNOW = WORK); LEARNING is the service that extracts discovery at scale.
 - **IDF_IS_ARCHETYPE** the Invention Disclosure Form is the archetype: structured records capturing what was discovered, when, by whom, with what evidence; LEARNING generalizes this to all scopes.
 - **EVERY_SCOPE_HAS_LEARNING_MD** every scope has LEARNING.md with the Date | Signal | Pattern | Source table; it is the compact governed form.
 - **GRADIENTS_BACKPROPAGATE** discovers backpropagate gradients to upstream scopes; LEDGER streams (TALK, CONTRIBUTE, EMAIL, PROVISION) are INTEL sources that each become a LEARNING row.
- ## Constraints

MUST: Compose INTEL primitive every discovery evidence

MUST: Follow governed record shape (generalized ID Form)

MUST: Source from governed operations only no fabrication

MUST: Backpropagate gradients to upstream scopes

MUST: Every scope has LEARNING.md the pattern table

MUST: LEARNING.md pattern table: Date | Signal | Pattern | Source

MUST: Ingest LEDGER streams TALK, CONTRIBUTE, EMAIL, PROVISION

MUST: Every ingested LEDGER entry becomes a LEARNING row

MUST NOT: Fabricate discoveries

MUST NOT: Store unstructured data outside governed record

MUST: GC frozen epochs after rotation, delete the rest

MUST NOT: Bypass provenance chain every gradient trace

MUST NOT: Ignore LEDGER streams silence is not governance

MUST: Flat service layout no singleton stage directories

MUST: CAS fanout git-style 2-char prefix buckets

MUST: Manifest sharding one shard per signal type

MUST: Incremental discovery checkpoint per repo, signal type

MUST NOT: Store 100K+ entries in a single manifest file

MUST NOT: Store 100K+ files in a single flat CAS directory

MUST: Consume patterns build reads LEARNING.md back

MUST: Surface regression signals repeated DEBIT:DISCOVERY

MUST: Surface growth signals repeated MINT:WORK

MUST: Feed INTEL consumed patterns wire into system

MUST NOT: Capture without consumption write-only LEARNING

MUST: Novelty gate candidates with <20% new keywords

MUST: Cross-scope propagation CROSS_INTEL targets

MUST: Emit LEARNING-METRICS.json promotion rate, signal type

MUST NOT: Pollute fallback scope unroutable patterns

MUST: Trivial filter shared TRIVIAL_PATTERNS (>=50)

MUST: Synthesis dedups boundary refs by hash BEFORE CAS resolution. BOUNDARIES shards overlap (o
MUST: Synthesis is idempotent under unchanged input. A BOUNDARIES fingerprint (mtime+size over B

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